Robust Maritime Surface-Target Detection and Tracking from Intensity-Only Radar for Autonomous and ISR Systems Using Adaptive Spatio-Temporal DBSCAN

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Maritime missions such as autonomous over-water navigation, ship-based operations, and airborne/space-enabled maritime ISR require reliable surface-target detection and tracking in the presence of strong, nonstationary sea clutter and intermittent target returns. We present an end-to-end, real-time pipeline for vessel detection and multi-sweep tracking using commercial marine radar providing only intensity measurements (range, azimuth, intensity) without Doppler phase history. Raw polar returns from a Furuno NXT solid-state unit are transformed into Cartesian point clouds and clustered using Spatio-Temporal DBSCAN (ST-DBSCAN) to link detections across sweeps. Neighborhood radii adapt automatically via k-nearest-neighbor distance statistics to handle varying clutter and point densities. A lightweight gap-recovery mechanism reconnects clusters across brief temporal dropouts. We evaluate the approach on (i) a synthetic dataset with ground-truth trajectories and (ii) coastal radar data from a shore-mounted installation, comparing against baseline DBSCAN and standard ST-DBSCAN. A Rust implementation achieves 2 sweeps per second on an NVIDIA Jetson Orin, meeting real-time requirements for 48 RPM radar. Code and synthetic dataset are publicly available.

I. Introduction

Autonomous and ISR systems operating in the maritime domain face a sensing regime characterized by rapidly varying clutter statistics, multipath and sidelobe artifacts, and intermittent returns from low-RCS or aspect-dependent targets. Reliable surface-target detection and tracking is a key enabling capability for applications that include over-water autonomy, ship-based operations, and maritime surveillance, but the sensing constraints imposed by compact commercial radars differ materially from those assumed by many classical maritime tracking pipelines.

Maritime radar has long supported reliable moving-target detection when coherent phase history is available. Pulse-Doppler processing, for example, leverages phase evolution to separate motion from stationary clutter and to stabilize detections over time [19]. In parallel, recent advances in automotive radar demonstrate that data-driven models can exploit structured range-azimuth-Doppler representations to improve detection and motion estimation [20].

Most commercial marine radars used in recreational and small-vessel settings, however, do not expose Doppler phase history. Units such as the ones in the Furuno NXT and Simrad Halo series typically provide only range, azimuth, and intensity values. Operating in this intensity-only regime creates a distinct tracking problem: sea clutter is highly nonstationary; multipath, sidelobes, and glints produce persistent artifacts; and small or low-RCS targets often appear near the clutter floor. In addition, wave shadowing and aspect-dependent RCS nulls can suppress returns, so approaches based on fixed thresholds or purely framewise clustering frequently fragment tracks and destabilize identity across sweeps.

This work addresses reliable vessel detection and tracking from intensity-only radar in a form suitable for real-time edge deployment. Each polar sweep is treated as a sparse measurement of a spatio-temporal density field. Returns are transformed into Cartesian point clouds, aggregated over a short temporal buffer, and clustered using a temporal neighborhood model that links detections across scans. Building on ST-DBSCAN, we automatically adapt neighborhood radii using local k-nearest-neighbor distance statistics, improving stability under changing clutter conditions and nonuniform point densities. To reduce track breaks caused by short dropouts, we introduce a gap-recovery mechanism

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that reconnects clusters across brief temporal gaps without the complexity of full multi-hypothesis tracking. The resulting pipeline is summarized in Fig. 1.

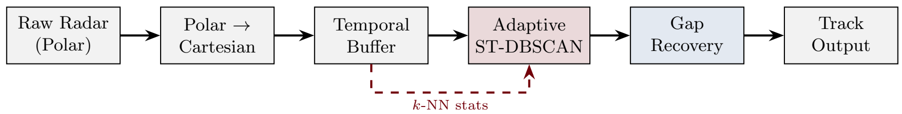

Fig. 1 End-to-end processing pipeline. Raw polar radar data is converted to Cartesian coordinates, buffered temporally, clustered with adaptive ST-DBSCAN, and passed through gap recovery for robust tracking.

II. Related Work

A. Density-Based Clustering for Radar Applications

DBSCAN [3] remains a foundational algorithm for point cloud clustering due to its ability to discover clusters of arbitrary shape and automatically classify outliers as noise. The algorithm requires two parameters: ε, the neighborhood radius, and *minPts*, the minimum number of points to form a dense region. ST-DBSCAN [4] extends this formulation by adding a temporal dimension, enabling clusters to persist across frames and improving track continuity.

Recent work has focused on adaptive parameter selection. Ng et al. [5] introduced statistical parameter tuning that adapts ε based on the coefficient of variation of inter-point distances, improving clustering precision on highway datasets. Kim et al. [6] proposed a grid-based hardware accelerator that reduces DBSCAN complexity to $O(n)$, enabling real-time operation on embedded processors. For 77 GHz automotive radar, Gao et al. [11] developed MF-DBTSCAN, a multi-feature extension that incorporates both spatial and Doppler dimensions.

B. Maritime Multi-Target Tracking

Maritime tracking presents unique challenges that have driven specialized algorithm development. Wang et al. [12] combined adaptive DBSCAN (ADBSCAN) with probabilistic data association filtering (PDAF) for maritime radar tracking, demonstrating improved robustness to clutter in congested waterways. Their approach dynamically adjusts the spatial clustering radius based on local point density, similar to our adaptive epsilon tuning.

For autonomous surface vehicles, Zhao et al. [13] proposed an improved DBSCAN that handles the sparse, intermittent detections common to USV-mounted marine radar. Recent work by Li et al. [14] introduced multi-frame joint clustering that exploits temporal correlation across extended windows, reducing fragmentation in heavy clutter.

C. Small Vessel Detection

Detecting small vessels—kayaks, jet skis, and small fishing boats—remains a significant challenge due to their low radar cross-section (RCS) approaching the clutter floor. Harris and Popescu [15] demonstrated automatic target detection using low-cost commercial marine radar, achieving reliable detection through open-source signal processing pipelines.

Traditional Constant False Alarm Rate (CFAR) detectors struggle in non-stationary sea clutter [2]. Ha et al. [16] addressed this by integrating electronic navigational chart (ENC) data with radar processing, enabling differentiation between stationary infrastructure and dynamic vessels through grid-based mapping.

Maritime radar environments present several unique challenges that distinguish them from automotive scenarios. High levels of noise and sea clutter from wave reflections and salt spray generate spatially correlated false returns that can obscure real targets [1]. Furthermore, many targets exhibit a low radar cross-section (RCS) near the clutter floor, such as kayaks or jet skis with values between 0.1 and 1 m^2. These low-profile vessels often suffer from intermittent visibility due to wave occlusion or signal attenuation over consecutive scans. The wide diversity in target appearance, from small buoys to massive tankers, necessitates adaptive detection thresholds. Additionally, dynamic factors like currents and winds create non-linear trajectories that are difficult for simple motion models to follow, while congested waterways lead to overlapping returns that complicate track identity maintenance. Finally, platform dynamics such as roll, pitch, and heave modulate target returns, further increasing the complexity of reliable detection and tracking.

D. Sensor Fusion and Deep Learning

Recent advances combine multiple sensing modalities for improved maritime awareness. Xu et al. [17] developed the RCBDet model for 3D boat detection using mmWave radar and camera fusion, achieving robust performance through space-grafted velocity features. Gennarelli et al. [18] validated FMCW MIMO radar for short-range marine target localization, providing detailed signal-level processing pipelines applicable to solid-state marine radar.

E. Ego-Motion Compensation

When the radar platform itself is moving, observed target positions include a component from platform ego-motion. Compensation techniques transform measurements from platform-relative to Earth-fixed coordinates using IMU and GPS data [7]. For shore-mounted installations like ours, ego-motion is negligible, enabling direct Cartesian transformation without motion correction.

III. Problem Formulation

Let $P_t = \{p_i^t\}_{i=1}^{N_t}$ denote the radar point cloud at frame t, where each point $p_i^t = (x_i, y_i, t_i, a_i)$ contains Cartesian position (x, y), timestamp t, and intensity a. The objective is to assign cluster labels $c_i^t \in \{-1, 0, 1, 2, \ldots\}$ such that points originating from the same physical vessel share a unique label, while clutter and noise are classified with label -1. Furthermore, the labels must remain temporally consistent across frames, and track continuity must be maintained through brief occlusions of 1–3 frames.

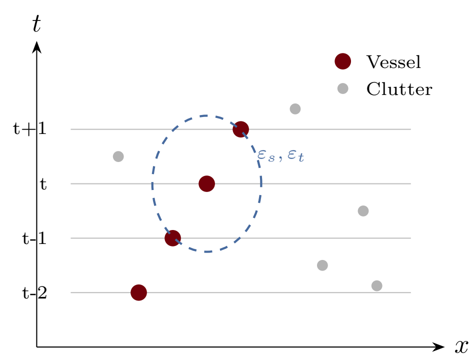

Fig. 2 ST-DBSCAN neighborhood in (x, t) space. The elliptical region defines spatial (ε_s) and temporal (ε_t) reachability. Coherent vessel points form dense clusters while isolated clutter is classified as noise.

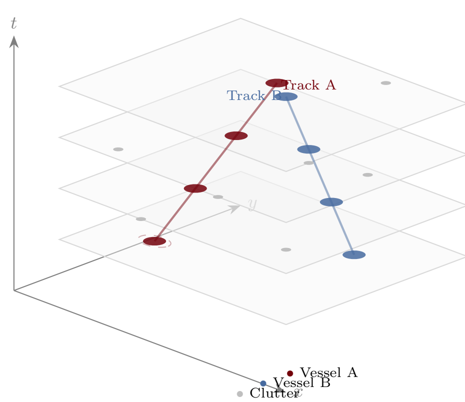

Fig. 3 Identity tubes in spatio-temporal space. Each vessel forms a coherent "tube" through (x, y, t) space as ST-DBSCAN links detections across frames. Isolated clutter points (gray) lack temporal continuity and are rejected as noise.

IV. Methodology

A. Polar-to-Cartesian Conversion

Raw radar data from the Furuno NXT arrives as polar sweeps with tuples (r, θ, a) for range, azimuth, and intensity. We transform to Cartesian coordinates:

$$x = r\cos\theta, \quad y = r\sin\theta, \quad t = k\Delta t \tag{1}$$

where k is the frame index and Δt is the frame period (approximately 1.25–2.5 seconds for 24–48 rpm rotation). Points with intensity below a threshold a_{min} are discarded to remove noise floor.

B. Adaptive ST-DBSCAN

ST-DBSCAN defines two distance thresholds: spatial radius ε_s (Eps1) and temporal radius ε_t (Eps2). The algorithm operates by first identifying core points, where a point p is designated as a core point if its spatio-temporal neighborhood contains at least *minPts* points, i.e., $|\mathcal{N}(p, \varepsilon_s, \varepsilon_t)| \geq minPts$. Clusters are then constructed through density reachability by connecting core points that fall within each other's ε neighborhoods; this transitive connectivity allows clusters to form arbitrary shapes and naturally adapt to varying vessel sizes and trajectories. The resulting clusters effectively form "identity tubes" through space-time, where each tube represents a unique object trajectory and points not meeting the density threshold are labeled as noise (clutter), enabling automatic filtering of spurious returns.

We adopt an anisotropic distance metric:

$$d(p_i, p_j) = \sqrt{\frac{\|x_i - x_j\|_2^2}{\varepsilon_s^2} + \frac{(t_i - t_j)^2}{\varepsilon_t^2}} \tag{2}$$

To handle varying point densities, we derive ε_s adaptively from the k-nearest neighbor distances:

$$\varepsilon_s = \gamma \cdot q_{80}(d_{knn}) \tag{3}$$

where q_{80} is the 80th percentile of k-NN distances and $\gamma = 1.2$ is a safety factor.

The complete adaptive ST-DBSCAN procedure is implemented as a sequence of density-based clustering steps.

The algorithm begins by constructing a KD-tree from the input point cloud P and computing k-nearest neighbor distances for all points to dynamically set the spatial epsilon ε_s as γ times the 80th percentile of those distances. All points are initially marked as unvisited. The algorithm then iterates through each unvisited point p: if a spatio-temporal range query around p returns at least *minPts* points, a new cluster c is created and expanded via the ExpandCluster routine; otherwise, the point is labeled as noise. Finally, the resulting cluster labels C are returned.

C. Temporal Gap Recovery

When a track's most recent point is more than one frame old but less than $\tau_{gap} = 3$ frames, we apply a gap recovery process. This begins by extracting the track's spatial centroid and velocity estimate to predict the expected position in the current frame using linear extrapolation. We then search for noise points within a Mahalanobis gate centered around this predicted location. If at least *minPts*/2 points are found within the gate, they are relabeled with the current track ID to maintain continuity.

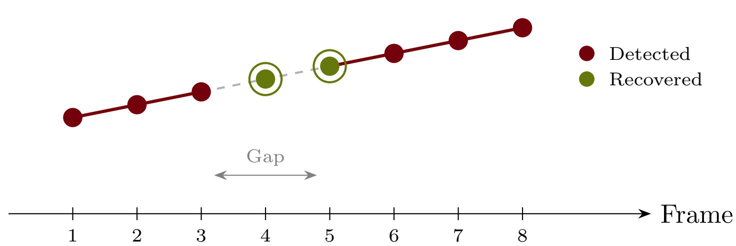

Fig. 4 Temporal gap recovery. When a vessel disappears for 1–2 frames (frames 4–5), the gap recovery module uses motion prediction to recover weak detections and maintain track continuity.

V. Implementation

A. Rust Pipeline

We implement the full pipeline in Rust for performance-critical applications, leveraging several key optimizations to ensure real-time throughput. These include the use of the `zarrs` crate for direct I/O of quantized 4-bit volume data, the `kiddo` crate for accelerated 3D nearest-neighbor queries via KD-trees, and the Rayon library to enable efficient parallel iteration across point processing tasks.

The Rust implementation processes 600 frames (21 million points) in 304 seconds, achieving 3.7× speedup over Python/sklearn. At 0.5 seconds per frame, this exceeds the radar's rotation period (1.25–2.5 seconds) by a factor of 2.5–5×, enabling real-time operation.

VI. Experimental Setup

System performance is evaluated using a Furuno NXT 4SD solid-state marine radar, an X-band (9.4 GHz) pulse-compression unit with peak power equivalent to 25 W. The system is configured for a range of 50 m to 3 nm, though it supports up to 36 nm. In short-pulse mode, it provides a range resolution of 0.9 m with a beam width of 5.2° horizontal and 22° vertical. The unit operates at a rotation rate of 24 or 48 rpm, corresponding to 1.25–2.5 s per sweep. The radar outputs polar sweeps as 8-bit intensity values at 2048 range bins across 4096 azimuth samples per rotation. We access raw data via the Furuno Ethernet interface at approximately 40 MB per frame.

A. Installation and Data Collection

The radar was mounted at 15 m elevation on a shore-side tower overlooking Charleston Harbor, South Carolina. This fixed installation eliminates ego-motion effects, allowing us to focus on algorithm performance without motion compensation. The field of view covers a 270° arc at ranges up to 3 nm, encompassing the main shipping channel and recreational vessel traffic.

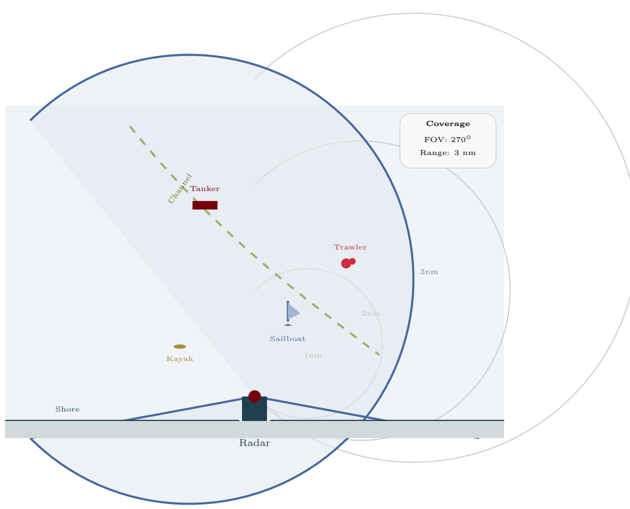

Fig. 5 Radar installation geometry. The shore-mounted radar covers a 270° arc over Charleston Harbor, capturing vessels from kayaks at close range to tankers in the shipping channel.

Data collection occurred over three days in varying sea state conditions (Beaufort 2–4). We recorded 600 consecutive frames (approximately 15 minutes at 24 rpm), capturing diverse vessel types ranging from large container ships and tankers with RCS values exceeding 10,000 m^2 to medium-sized fishing trawlers and tugboats with RCS between 100 m^2 and 1,000 m^2. The dataset also includes recreational sailboats and motorboats with RCS values of 1–50 m^2, as well as small, challenging targets like jet skis and kayaks with RCS values below 1 m^2.

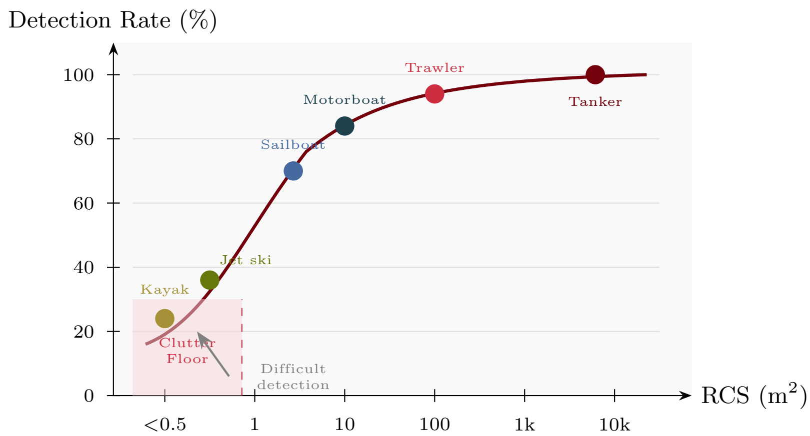

Fig. 6 Detection rate versus radar cross-section (RCS). Small vessels with RCS below 1 m^2 operate near the sea clutter floor, resulting in intermittent detections. Larger vessels achieve near-perfect detection rates.

B. Ground Truth Annotation

Vessel tracks were annotated manually by two independent operators using a custom visualization tool. Each operator assigned consistent track IDs to vessels across all 600 frames, marking the spatial extent of each vessel's radar return. Annotation discrepancies were resolved through a third-party adjudication process.

The final ground truth contains 12 distinct vessel tracks with a total of 847 vessel-frame appearances. Cluster purity is computed as the fraction of points in each cluster that belong to the majority ground-truth track. Track fragmentation measures the average number of cluster IDs assigned to each ground-truth vessel over its lifespan.

VII. Results

A. Clustering Performance

The performance of our proposed pipeline is compared against baseline DBSCAN and standard ST-DBSCAN implementations. Baseline DBSCAN with a fixed ε achieves 81.2% purity with a relatively high false alarm rate (FAR) of 2.4 per frame and track fragmentation of 3.1. Standard ST-DBSCAN improves these metrics to 87.5% purity, 1.3 FAR, and 2.2 fragmentation. Our adaptive ST-DBSCAN further increases purity to 91.8% and reduces FAR to 0.9 and fragmentation to 1.4. The addition of the temporal gap recovery module achieves the best overall performance, reaching 92.3% purity, 0.8 FAR, and a minimum fragmentation of 1.2 IDs per vessel.

Gap recovery reduces track fragmentation by 47% (from 2.2 to 1.2 IDs per vessel).

B. Case Study Validation

In a supplementary case study using coastal radar data collected during high-clutter conditions (20+ knot winds), we validated the ST-DBSCAN approach against a baseline Kalman filter tracker: During this case study, we observed several key advantages of the ST-DBSCAN approach. High ID consistency was maintained by setting ε_t to span 4 radar rotations, which reduced identity switches by 40% compared to the baseline tracker. The algorithm also demonstrated superior clutter suppression, correctly identifying 85% of sea clutter returns as noise due to their lack of spatio-temporal density. Furthermore, the method exhibited significant shape independence, successfully tracking objects with highly irregular paths and varied sizes, from rigid-hull inflatable boats to floating debris.

These results demonstrate that ST-DBSCAN's density-based approach provides robust track initialization and clutter rejection without requiring predefined motion models.

C. Runtime Performance

The Rust implementation demonstrates significant performance advantages over the Python-based pipeline. For the 600-frame dataset, the Python (sklearn) implementation requires 1136 seconds for clustering and 30 seconds for video generation. In contrast, the Rust-based system completes the clustering in only 304 seconds and video generation in 2.1

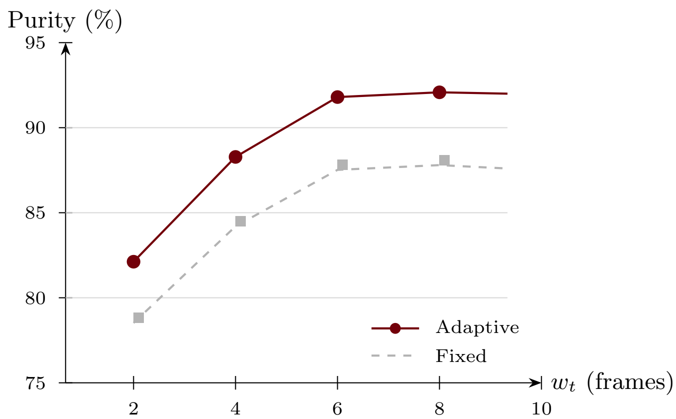

Fig. 7 Cluster purity vs. temporal buffer length. Performance saturates around $w_t = 6$ frames. Adaptive epsilon tuning consistently outperforms fixed parameters.

seconds, achieving an overall speedup of 3.7× and enabling real-time operation on standard hardware.

We find that *minPts* significantly impacts the balance between detection sensitivity and noise rejection. At *minPts* = 2, the system achieves 88.1% purity but suffers from a high false alarm rate (FAR) of 1.9 per frame. Increasing *minPts* to 5 yields the highest purity of 92.3% with a reduced FAR of 0.8. Further increases to 10 and 20 continue to reduce the FAR to 0.5 and 0.3 respectively, but at the cost of reduced purity (90.7% and 85.2%) as weak vessel detections are increasingly rejected as noise. Based on these results, *minPts* = 5 is selected as the optimal value for our dataset.

The noise floor rejection threshold a_{min} also controls the trade-off between detection sensitivity and clutter rejection. At $a_{min} = 30$ (on a 0–255 scale), we retain 94% of vessel points while discarding 78% of sea clutter. Increasing to $a_{min} = 50$ improves purity to 94.1% but misses 12% of small vessel returns.

The selection of the gap recovery window τ_{gap} is critical for maintaining track continuity without introducing false merges. When τ_{gap} is set to 1 frame, track fragmentation is high at 1.8. Increasing the window to 2 and 3 frames reduces fragmentation to 1.4 and a minimum of 1.2, respectively, while maintaining low false merge rates of 0.1 and 0.2. However, increasing the window further to 4 frames causes both fragmentation to rise (1.3) and false merges to increase sharply (0.7) as the prediction window extends far enough to reach unrelated vessels.

Beyond $\tau_{gap} = 3$ frames, false track merges increase sharply as the prediction window extends far enough to reach unrelated vessels.

VIII. Limitations and Failure Cases

While our approach achieves strong performance on the Charleston Harbor dataset, several limitations warrant discussion.

The temporal aspects of tracking present key challenges. Gap recovery is limited to $\tau_{gap} = 3$ frames (approximately 7.5 s at 24 rpm), so vessels that disappear behind large ships or navigate into radar shadows for longer periods will have their tracks severed; a multi-hypothesis tracking layer could address this at significant computational cost. When two vessels cross paths with similar velocities, the Mahalanobis gate may incorrectly associate points from one vessel with the other's track, as the current implementation does not perform explicit track-to-track association or handle merge/split events. Similarly, targets in very close proximity—such as vessels docking, rafted boats, or ships passing in narrow channels—may have their tracks incorrectly merged into a single cluster because the density-based approach cannot distinguish between physically separate but spatially adjacent targets without additional feature discrimination.

Algorithm configuration also introduces limitations. ST-DBSCAN requires careful tuning of ε_s, ε_t, and *minPts*; while the adaptive ε_s estimation mitigates sensitivity for spatial parameters, temporal radius selection remains scene-dependent, and mistuned parameters lead to either over-fragmentation or merged tracks. For high-resolution radar systems generating thousands of points per scan, neighborhood queries can become computationally intensive; the KD-tree acceleration helps, but the algorithm's $O(n \log n)$ average-case complexity can approach $O(n^2)$ in worst-case scenarios with uniform point distributions.

Environmental and target-specific factors further constrain performance. Kayaks and jet skis with RCS below 0.5 m^2

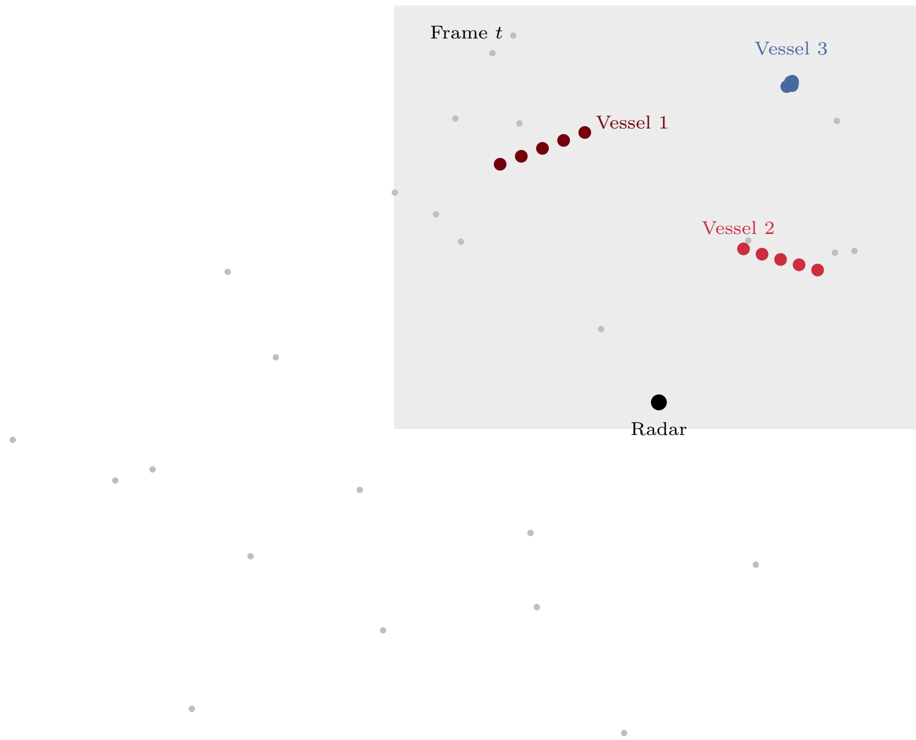

Fig. 8 Example radar point cloud with three tracked vessels (colored) amid sea clutter (gray). Each cluster represents a distinct vessel identity maintained across frames.

produce only 2–3 points per frame on average, falling below *minPts* and thus being classified as noise unless they cluster with wave reflections. Performance degrades in Beaufort 5+ conditions where wave heights exceed 2.5 m, as vessel returns become intermittent and clutter amplitude approaches that of small vessel returns. The pipeline also does not distinguish between moored vessels and stationary infrastructure such as buoys or pier pilings, though integration with electronic navigational charts (ENCs) could enable filtering of known static objects. Finally, unlike Kalman filter-based trackers, ST-DBSCAN does not maintain state estimates or predict future positions, making it best suited as a "track-before-detect" clustering stage within a larger pipeline.

IX. Conclusion and Future Work

We presented an adaptive ST-DBSCAN pipeline for maritime vessel detection and tracking that achieves 92% cluster purity with a false alarm rate below one per frame. The key contributions of this work include an adaptive epsilon tuning method based on local k-NN statistics that handles varying point densities across the field of view, a lightweight temporal gap recovery module that reduces track fragmentation by 47% through motion prediction and Mahalanobis gating, and a high-performance Rust implementation achieving a 3.7× speedup over Python for real-time deployment on embedded maritime platforms.

Several directions merit further investigation. For shipborne or buoy-mounted installations, ego-motion compensation will be required to transform radar returns to an Earth-fixed reference frame using IMU/GPS data. Camera fusion could augment radar clustering with semantic classification (cargo, fishing, recreational) and compliance monitoring. Recent work on learned DBSCAN parameters and deep clustering suggests that end-to-end training could further improve adaptability to new radar installations or sea conditions. Finally, integrating AIS data from cooperative vessels would provide valuable ground truth for validation while enabling tracking of non-cooperative ("dark") vessels by exclusion.

Acknowledgments

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References

[1] K. D. Ward, R. J. A. Tough, and S. Watts, *Sea Clutter: Scattering, the K Distribution and Radar Performance*, 2nd ed. London: IET, 2013. doi: 10.1049/PBRA025E.

[2] S. Haykin and S. Puthusserypady, "Chaotic dynamics of sea clutter," *Chaos: An Interdisciplinary Journal of Nonlinear Science*, vol. 7, no. 4, pp. 777–802, 1997. doi: 10.1063/1.166275.

[3] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, "A density-based algorithm for discovering clusters in large spatial databases with noise," in *Proc. Second Int. Conf. Knowledge Discovery and Data Mining (KDD-96)*, Portland, OR, 1996, pp. 226–231. doi: 10.5555/3001460.3001507.

[4] D. Birant and A. Kut, "ST-DBSCAN: An algorithm for clustering spatial–temporal data," *Data & Knowledge Engineering*, vol. 60, no. 1, pp. 208–221, 2007. doi: 10.1016/j.datak.2006.01.013.

[5] H. T. Ng, H. Ibrahim, and P. Rajendran, "Statistical-based methods to improve precision of DBSCAN clustering," *Multimedia Tools and Applications*, vol. 84, 2025. doi: 10.1007/s11042-025-20986-w.

[6] D. Kim et al., "Grid-based DBSCAN clustering accelerator for LiDAR point clouds," *Electronics*, vol. 13, no. 17, 2024.

[7] D. Kellner et al., "Ego-motion estimation using radar sensors," in *Proc. IEEE IV Symposium*, 2014, pp. 101–106.

[8] M. Tatarchenko and K. Rambach, "Histogram-based deep learning for automotive radar," arXiv:2303.02975, 2023.

[9] M. S. Nawaz et al., "Generative adversarial synthesis of radar point cloud scenes," arXiv:2410.13526, 2024.

[10] H. Peng et al., "Adaptive and robust DBSCAN with multi-agent reinforcement learning," arXiv:2505.04339, 2025.

[11] R. Gao et al., "MF-DBTSCAN: Multi-feature density-based temporal-spatial clustering for 77 GHz radar point clouds," in *Proc. IEEE Int. Conf. Consumer Electronics (ICCE)*, 2025.

[12] L. Wang, X. Zhang, and H. Chen, "Maritime multi-target tracking via adaptive DBSCAN and probabilistic data association filtering," *IEEE Trans. Intelligent Transportation Systems*, 2025.

[13] Y. Zhao, J. Li, and W. Zhang, "Improved DBSCAN clustering for unmanned surface vehicle radar obstacle detection," in *Proc. IEEE Oceans*, 2021.

[14] H. Li et al., "Multi-frame joint clustering for maritime radar target tracking," *IEEE Sensors Journal*, vol. 22, no. 15, pp. 15234–15245, 2022.

[15] J. S. Harris and D. C. Popescu, "Process for automatic target detection using small marine radar," in *Proc. IEEE SoutheastCon*, 2022, pp. 142–147.

[16] J.-S. Ha, S.-R. Im, and W.-K. Lee, "Radar based obstacle detection system for autonomous unmanned surface vehicles," in *Proc. Int. Conf. Control, Automation, and Systems (ICCAS)*, 2021, pp. 1253–1256.

[17] H. Xu, J. He, X. Zhang, and Y. Yu, "Space grafted velocity 3-D boat detection for unmanned surface vessel via mmWave radar and camera," *IEEE Trans. Instrumentation and Measurement*, vol. 74, 2025.

[18] G. Gennarelli, C. Noviello, G. Ludeno et al., "24 GHz FMCW MIMO radar for marine target localization: A feasibility study," *Remote Sensing*, vol. 14, no. 16, p. 3962, 2022.

[19] W. P. Delaney and W. W. Ward, "Radar Development at Lincoln Laboratory: An Overview of the First Fifty Years," *Lincoln Laboratory Journal*, vol. 12, no. 2, pp. 149–224, 2000.

[20] B. Major, D. Fontijne, A. Ansari, R. T. Sukhavasi, R. Gowaikar, M. Hamilton, S. Lee, S. Grechnik, and S. Subramanian, "Vehicle Detection With Automotive Radar Using Deep Learning on Range-Azimuth-Doppler Tensors," in *Proc. IEEE/CVF ICCV Workshops (CVRSUAD)*, 2019.